9.11 Writing and Solving Quadratic Equations

Lesson Introduction

Benchmarks	MA.912.AR.3.1 Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real number system. MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.	Learning Target	• I can write and solve one-variable quadratic equations.

	Building On	Working Toward
Benchmark Coherence	MA.8.AR.2.3 Given an equation in the form of $x^2 = p$ and $x^3 = q$, where p is a whole number and q is an integer, determine the real solutions.	MA.912.AR.3.2Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real and complex number systems.

Essential Question	• How can I write and solve a quadratic equation given a real-world context?
Lesson Narrative	In this lesson, students will apply all they have learned in Unit 11 to write and solve quadratic equations given real-world scenarios.

Component Summary	9.11.1 Warm-Up (~5 min.)	Rubber duck pattern activity *(SE p. 114)*	9.11.3 Guided Practice (15 min.)	Writing and solving quadratic equations practice *(SE p. 117)*
	9.11.2 Guided Instruction (15 min.)	Writing and solving quadratic equations activity *(SE p. 115)*	9.11.4 Your Turn! (10 min.)	Writing and solving quadratic equations independent practice *(SE p. 118)*
	9.11.5 Wrap-Up (~5 min.)	Self-reflection on understanding of unit content *(SE p. 118)*		

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms:</u> N/A <u>Additional Terms:</u> • Constraints • Quadratic equation • Vertex form • Standard form	• N/A	• Work out each question prior to beginning the lesson. Identify potential misconceptions that may arise based on the context provided in each word problem. Look for ways to address misconceptions your students have had throughout the unit. Look for ways to continue spiraling content by asking questions about key features and their interpretations.

<table>
<tr><td rowspan="4">Teacher Tips</td><th>Common Misconceptions</th><th>Things to Consider</th></tr>
<tr><td>Students may misinterpret the key information from the word problems.
Students may not translate the word problems to quadratic equations correctly.
Students may make calculation errors.</td><td>Students may need support with deciphering word problems. Consider leveraging a word problem strategy to decipher the key information from each problem and how to approach them.</td></tr>
<tr><th>Literacy and Language Routines</th><th>Mathematical Thinking and Reasoning (MTR) Standard</th></tr>
<tr><td>Three Reads
This lesson consists primarily of word problems that require students to read extensively. Consider leveraging a three-reads strategy. In the first read, ensure that students understand the context. In the second read, have students interpret the question. In the third read, have students identify important information.</td><td>MA.K12.MTR.3.1: Mathematical Fluency
Students will be working with various contexts that represent quadratic equations. They must choose the most appropriate method for solving a quadratic equation based on the given context.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">9.11.2 Guided Instruction provides students with a collaborative experience while discussing solutions to problems. Student performance on these questions can pinpoint potential differentiation scenarios.
Consider using visual highlighting techniques for word problems to assist visual learners with identifying key information in the problem.
The 9.11.4 Your Turn! provides an opportunity for students to write and solve a quadratic equation given a word problem. Student performance on these two questions can provide valuable data for differentiation.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

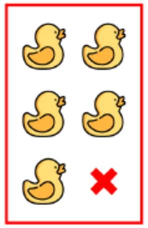
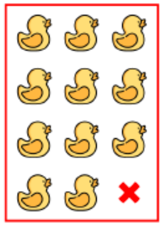
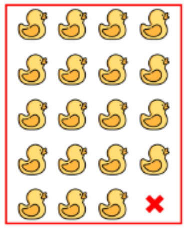
9.11.1 Warm-Up: Rubber Duck Quadratics

Component Overview: *Students will explore a rubber duck pattern. In this activity, students will describe the pattern using visual representations and expressions.*

9.11.1 Warm-Up: Rubber Duck Quadratics

A pattern made of rubber ducks is shown.

1. Describe the pattern.
   Answers will vary. The pattern shows a number squared plus the number minus one. The pattern shows a multiplication of two numbers minus one.
2. To describe the pattern, Emily made drawings and created an equation. Her drawings are shown below. Her equation is $D = (N + 1)(N + 2) - 1$.

$N = 1$	$N = 2$	$N = 3$
		

Use Emily's drawing for $N = 3$ to show how each expression from the equation $D = (N + 1)(N + 2) - 1$ is illustrated in her drawing.

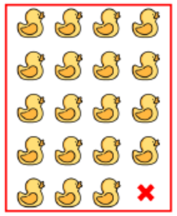
Answers will vary. The top row is $(N + 1)$ because $(3 + 1) = 4$. The left column is $(N + 2)$ because $(3 + 2) = 5$. Then, $4 \times 5 = 20$. The -1 represents the x, leaving a total of 19 ducks.

Expressions

$(N + 1)$

$(N + 2)$

-1



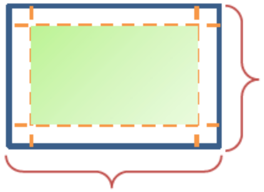
Allow students to share their thoughts on the pattern and their answers to this Warm-Up. Students that were unable to identify the pattern may understand it more if they hear it being described by their peers.

9.11.2 Guided Instruction: Writing and Solving Quadratic Equations

Component Overview: *Students will receive guided instruction on writing and solving quadratic equations given a real-world context.*

9.11.2 Guided Instruction: Writing and Solving Quadratic Equations

1. A company is experimenting with the size of a gift box without a top that can be made from a 12 inch (in.) by 24 in. piece of cardboard. The company wants to know if the height, x, of the box is changed, what the area of the bottom of the box will be.



   a. What is the width of the bottom of the box?
      $12 - 2x$

   b. What is the length of the bottom of the box?
      $24 - 2x$

   c. Discuss with your partner the constraints on the value of x, the height of the box.
      Answers will vary. The height of the box, x, cannot be 6 or larger because if it is 6 or larger, the width would be zero or a negative number.

   d. Write the function A that can be used to determine the area, in square inches, in.^{2}, of the bottom of the box as a function of the height of the box, x.
      $A(x) = (12 - 2x)(24 - 2x)$

Consider applying a word problem strategy as you open this activity. For examples of word problem strategies, refer to the Unit Guide for this unit. Assist students with setting the stage for the context of this question to make a smooth transition through the associated questions and tasks.

For students who may be struggling with this concept, consider using the time in which students are released to work on each part as an opportunity to provide support to those students. The questions in this section are scaffolded in such a way that it can lead students to conceptual understanding. With additional support, struggling students can connect to the word problem.

Have students share their constraints with the class. Poll the class to see who agrees and disagrees with justification.

9.11.2 Guided Instruction: Writing and Solving Quadratic Equations (continued)

e. What is the height of the box if the area of the bottom of the box is 189 in.[2]?

$189 = (12 - 2x)(24 - 2x)$

$189 = 288 - 72x + 4x^2$

$0 = 4x^2 - 72x + 99$

$x = \dfrac{-(-72) \pm \sqrt{(-72)^2 - 4(4)(99)}}{2(4)}$

$x = \dfrac{72 \pm 60}{8}$

$x = 16.5 \qquad x = 1.5$

16.5 *inches or* 1.5 *inches*

f. Discuss with your partner whether the solutions are reasonable for the given context.

Answers will vary. The height of the box cannot be 16.5 *inches because one of the sides is only* 12 *inches. One of the constraints of the context was that the height of the box has to be less than* 6 *inches.*

Continue to re-emphasize word problem strategies during each question. Students will need additional practice with the deciphering skills.

2. A mathematician for a music company tracked the sales, in tens of millions of dollars, of music cassette tape, y, for x years since 1980 for the years 1980 to 2000. A table of values of some of the values is shown.

x	0	1	3	6	9	10	13	17	19
y	83	134	218	299	326	323	278	134	26

a. In what year did the sales of music cassette tapes reach its highest value?

In 1989

b. Work with your partner to explain how the table can be used to confirm when the sales reached its highest value.

Answers will vary. The table can show where the y-value changes from increasing to decreasing. This change represents the location of the vertex which is the maximum/minimum. The table can be used to find x-values for which the y-values are the same.

Each question provides an opportunity for specifying differentiation and support may be needed. Possible scenarios:

1. *Understanding data from a table*
2. *Choosing an appropriate method for solving quadratics*
3. *Properly understanding information from a word problem*
4. *Inputting values appropriately*
5. *Evaluating*

9.11.2 Guided Instruction: Writing and Solving Quadratic Equations (continued)

c. Write an equation that models the context.

$y = a(x-9)^2 + 326$
$83 = a(0-9)^2 + 326$
$-243 = a(81)$
$a = -3$
$y = -3(x-9)^2 + 326$

d. In what year(s) were the sales 185 tens of millions of dollars?

$185 = -3(x-9)^2 + 326$
$185 = -3(x-9)^2 + 326$
$-141 = -3(x-9)^2$
$47 = (x-9)^2$
$\sqrt{47} = \sqrt{(x-9)^2}$
$\pm\sqrt{47} = x - 9$
$x = 9 \pm \sqrt{47}$
$x \approx 15.856 \quad x \approx 2.144$
In 1982 and 1995

- *What variable represents the sales in tens of millions of dollars?*
- *What variable will you be solving for?*

3. Mangrove forests are important habitats for fish. An ichthyologist finds that some fish use mangrove forests only during high tide. The ichthyologist label fish that only use mangrove forests during high tide as "high-tide users." The ichthyologist gives values between 0 and 6.25 to represent the frequency of occurrence high-tide users are found in the mangrove forests. When the tide has been rising for 6.1 hours, the frequency of occurrence reaches a maximum of 6.25. The frequency of occurrence is 5 when the high tide has been rising for 3.6 hours. The frequency of occurrence reaches 5 again after 8.6 hours.

a. Define the variables.

x represents the time in hours
y represents the frequency of occurrence

b. Write an equation, in vertex form, that models the context.

$y = a(x-6.1)^2 + 6.25$
$5 = a(3.6-6.1)^2 + 6.25$
$-1.25 = a(-2.5)^2$
$-1.25 = a(6.25)$
$a = -\frac{1}{5}$
$y = -\frac{1}{5}(x-6.1)^2 + 6.25$

c. Determine when the frequency of occurrence will be 3.75.

$3.75 = -\frac{1}{5}(x-6.1)^2 + 6.25$

$-2.5 = -\frac{1}{5}(x-6.1)^2$

$12.5 = (x-6.1)^2$

$\sqrt{12.5} = \sqrt{(x-6.1)^2}$

$\pm\sqrt{12.5} = x - 6.1$

$x = 6.1 \pm \sqrt{12.5}$

$x = 6.1 \pm 3.536$

$x \approx 2.564 \qquad x \approx 9.636$

The frequency of occurrence will be 3.75 at 2.564 hours and 9.636 hours.

?

What information from the context given represents the vertex?

9.11.3 Guided Practice: Writing and Solving Quadratic Equations

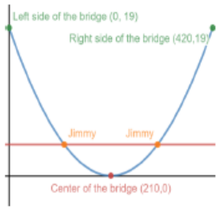
Component Overview: *Students will receive guided practice on writing and solving quadratic equations. Prior to the questions, read and review the context of the question out loud, including the notation of the parabola in context.*

9.11.3 Guided Practice: Writing and Solving Quadratic Equations

The Hal Adams bridge is a suspension bridge that crosses the Suwanee River near Mayo, Florida. The suspension wires are linked into and form a parabola, as shown in the photo of the bridge.
The lowest part of the parabola is located at the midpoint of the main span of the bridge, which is 420 feet long. The highest points of the suspension are 19 feet above the lowest point. The graph illustrates the parabola.

1. Write an equation that models the context. Leave the a-value as an unsimplified fraction.

$$y = a(x-210)^2$$
$$19 = a(0-210)^2$$
$$19 = a(-210)^2$$
$$a = \frac{19}{(-210)^2}$$
$$y = \frac{19}{(-210)^2}(x-210)^2$$



2. Jimmy is 4 feet tall. Where on the bridge will Jimmy be just as tall as the suspension cables?

$$4 = \frac{19}{44100}(x-210)^2$$
$$\frac{176400}{19} = (x-210)^2$$
$$\sqrt{\frac{176400}{19}} = \sqrt{(x-210)^2}$$
$$\pm\frac{420}{\sqrt{19}} = x-210$$
$$x = 210 \pm \frac{420}{\sqrt{19}}$$
$$x \approx 306.355 \qquad x \approx 113.645$$

Jimmy will be just as tall as the suspension cables at 113.645 ft and 306.355 ft from the left side of the bridge.

Continue to re-emphasize word problem strategies with each question. Students will need additional practice with the deciphering skills.

How did you determine the form of the equation that best models the context?

What does the vertex of the parabola in question 2 represent in the context?

9.11.4 Your Turn!

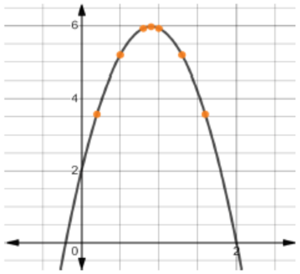
Component Overview: *Students will practice with writing and solving quadratic equations. This activity gives students practice and is designed to be done independently.*

9.11.4 Your Turn!

Matias launches a water balloon from a deck that is 2 feet off the ground. The time, in seconds (sec), from the launch and the height, in meters (m), of the water balloon is given in the table.

t (sec)	0.2	0.5	0.8	0.9	1	1.3	1.6
h (m)	3.568	5.185	5.92	5.969	5.92	5.185	3.568

A graph of the height of the water balloon is shown.



1. Write the function, $h(t)$, that models the height, h, in meters as a function of time, t, in seconds.
$h(t) = a(t - 0.9)^2 + 5.969$
$5.92 = a(1 - 0.9)^2 + 5.969$
$-.049 = a(0.1)^2$
$-0.049 = a(0.01)$
$a = -4.9$
$h(t) = -4.9(t - 0.9)^2 + 5.969$

2. Determine when the water balloon will hit the ground.
$0 = -4.9(t - 0.9)^2 + 5.969$
$-5.969 = -4.9(t - 0.9)^2$
$\frac{-5.969}{-4.9} = (t - 0.9)^2$
$\pm\sqrt{\frac{5969}{490}} = t - 0.9$
$t = 0.9 \pm \sqrt{\frac{5969}{490}}$
$t \approx 2.004 \qquad t \approx -0.204$
The water balloon will hit the ground in about 2 seconds.

Continue to re-emphasize word problem strategies with each question. Students will need additional practice with the deciphering skills.

Student performance on these two questions can provide valuable data for differentiation. This lesson does not have a Ticket Out the Door, making this activity the last in-text practice opportunity for students to show their level of understanding and mastery of the concepts of this lesson.

Why is the negative value not viable in this context?

9.11.5 Wrap-Up: Reflection

Component Overview: *Students will complete a unit reflection on their current level of understanding of this unit's concepts. This activity is designed to be done independently to provide accurate student self-reflection.*

9.11.5 Wrap-Up: Reflection

During the unit, you learned about writing and solving quadratics. Draw an **X** on each line to indicate your current level of understanding related to each concept.
Answers will vary.

1. Given the x-intercepts and another point on the graph of a quadratic function, write the equation for the function.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

2. Write a quadratic function to represent the relationship between two quantities from a graph, a written description, or a table of values within a mathematical or real-world context.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

3. Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real number system.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

4. Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

This Wrap-Up provides a student self-assessment. This data, coupled with their performance throughout the unit, can provide valuable insight into potential differentiation that may be necessary.